1. For $f = u + iv : \mathbb{C} \to \mathbb{C}$, denote $F = (u, v) : \mathbb{R}^2 \to \mathbb{R}^2$. Show that if f is holomorphic, then J_F is a $\mathbb{C}$-linear map from $\mathbb{C}$ to $\mathbb{C}$ and the Cauchy-Riemann equations are equivalent to

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

2. Show that the Cauchy-Riemann equations for $f = u + iv$ in the polar coordinates (r, θ) take the form:

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

3. Show that $\Delta = 4\frac{\partial^2}{\partial z \partial \bar{z}}$.

4. Let $T(z) = (az + b)/(cz + d)$ be a fractional linear transformation in normalized form. Assume that T has a fixed point z_1, with $z_1 \neq \infty$. Set $\kappa := T'(z_1)$, called the *multiplier* of T. We say that T is *parabolic* if $\kappa = 1$, *hyperbolic* if $\kappa > 0$ and $\kappa \neq 1$, *elliptic* if $|\kappa| = 1$ and $\kappa \neq 1$, and *loxodromic* otherwise. Prove that T is parabolic if and only if $a + d = \pm 2$, hyperbolic if and only if $a + d$ is real with $|a + d| > 2$, elliptic if and only if $a + d$ is real with $|a + d| < 2$, and loxodromic if and only if $a + d$ is non-real.
Hint. First, show that $\kappa = 1$ if and only if T has only one fixed point. Second, show that $\kappa = 1/f'(z_2)$ if z_2 is another finite fixed point of T. Third, show that $\kappa + 1/\kappa = (a + d)^2 - 2$.

5. Let f be a holomorphic function. The *Schwarzian derivative* of f is the expression

$$\mathcal{S}(f) = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2.$$

Show that
1) $\mathcal{S}(f) \equiv 0$ if and only if f is a fractional linear transformation.
Hint. Denote $\mathcal{N}(f) := (\log f')' = f''/f'$. Then, $\mathcal{S}(f) = \mathcal{N}(f)' - \frac{1}{2}\mathcal{N}(f)^2$.
2) For g holomorphic, $\mathcal{S}(g \circ f) = (\mathcal{S}(g) \circ f)(f')^2 + \mathcal{S}(f)$. In particular, if g is a fractional linear transformation, then $\mathcal{S}(g \circ f) = \mathcal{S}(f)$.